The GST law stack: complete derivations of the provable layer, and the remaining theorem stated precisely (Round 25 — the operator's order: prove the laws; the 22–23 all-arity law as the spine; the rotation; finish)

GST Hodge Campaign — Round 25 (session clock 2026-10-31)

2026-10-31 (session clock)

Disposition (on the face)

The operator's order: the GST laws that the closure uses are themselves unproven and must be proven; the operator's own 22–23 idea (the all-arity cyclic fold, recorded as the 2026-10-22 cyclic-original-X law and the single-factor Gaussian rotation) must be the spine, not a spectator; the rotation receives the actual mathematics; the work must end at the closing string.

This artifact does the part of that order that mathematics permits, and records the part it does not permit, with the receipts on the face:

1. Every GST law of the stack that admits a proof from first principles now has one, written out by hand in this file, in complete form, with no appeal to a certificate for any step of any proof: the rotation deck laws R1, R2, R5 (§2), the C_2 swap law (§2.5), the operator's all-arity cyclic law in full — the freeness lemma, the transfer theorem, the D_m mixed-Hodge computation, the Y_m mixed law, the S_m sector formula, the descent, the return leg $m \cdot \alpha$, the cyclotomic engine (§3), and the classical layer: Lefschetz $(1,1)$ in full, the hard-Lefschetz reduction, Lemma N the Newton conductor (§4).

2. The laws that do not admit a proof at the present state of the campaign are stated as THE REMAINING THEOREM (§6), each item in plain

mathematical terms with its receipt, together with the equivalence theorem: discharging the remaining theorem *is* proving the rational Hodge conjecture on the open sector. There is no third thing.

3. The closing string is not emitted. The one previous emission of it (Round 22) rested on Theorem D4(3), which the independent audit `AUDIT_22.md` found mathematically false — two computed counterexamples, one of them reproduced in §7 — and the claim was withdrawn from all artifacts. Emitting it again would re-assert a computed-false step. The operator's own audit loop (§36) is the mechanism that caught it; this round does not override that mechanism.

Each theorem below carries one of four labels, and the label is the claim's entire scope: **[PROVEN HERE]** (complete proof in this file), **[STANDARD STRENGTH]** (proof in this file modulo named standard theorems, each cited at textbook strength), **[CERTIFICATE RECEIPT]** (the campaign's committed exact-arithmetic check, cited; not load-bearing for any [PROVEN HERE] claim), **[OPEN]** (part of the remaining theorem; no proof exists).

1 The setting and the target

The target is the rational Hodge conjecture: for every connected smooth projective complex variety X and every rational (p,p) class $\alpha \in H^{2p}(X,\mathbb{Q})$, there exist algebraic cycles $Z_1,\dots,Z_r$ on X and rationals q_i with $\mathrm{cl}_X^{\mathrm{can}}(\sum_i q_i Z_i) = \alpha$.

The campaign's committed sector partition (recorded `2026-10-31-universal-close`, STEP 2; confirmed by the audits): $p = 1$ is Lefschetz $(1,1)$ (proven in full in §4.1 below); $p > d/2$ reduces to codimension $d-p \le d/2$ by hard Lefschetz (§4.2); $p \in \{0, d, d-1\}$ are the classical boundaries; the open sector is $2 \le p \le \lfloor d/2 \rfloor$, $d \ge 4$.

The campaign's committed world model: the eight-letter Gaussian lattice world on A with the Weil rays $W_R, W_I = \mathrm{Re}, \mathrm{Im}\,\Omega$, $\Omega = \prod_{j=0}^{3}(e_j + i f_j)$, the polarization $\theta = \sum_j e_j f_j$, the door pins $m^*(A,2,W_R) = m^*(A,2,W_I) = 2$ exactly, and the infinite census family $\{S_m(A)\}_{m\ge 2}$ of §3.

2 The rotation deck laws — complete proofs

2.1 Cyclic group cohomology at standard strength

Let $G = C_m = \langle\sigma\rangle$ act on a $\mathbb{Z}$-module M, and let $N = 1 + \sigma + \cdots + \sigma^{m-1}$ be the norm. The standard free resolution of $\mathbb{Z}$ over $\mathbb{Z}[G]$ is the periodic resolution

$$\cdots \to \mathbb{Z}[G] \xrightarrow{\sigma-1} \mathbb{Z}[G] \xrightarrow{N} \mathbb{Z}[G] \xrightarrow{\sigma-1} \mathbb{Z}[G] \to \mathbb{Z} \to 0$$

(exactness at each step is a two-line check: $\ker(\sigma-1) = \mathbb{Z}[G]N$ and $\ker N = \mathbb{Z}[G](\sigma-1)$; textbook strength, Brown, *Cohomology of Groups*, III.1). Applying $\mathrm{Hom}_{\mathbb{Z}[G]}(-, M)$ gives the cochain complex

$$M \xrightarrow{\sigma-1} M \xrightarrow{N} M \xrightarrow{\sigma-1} M \to \cdots,$$

whence

$$H^2(C_m; M) = \ker(\sigma-1)/NM = M^{C_m}/NM, \qquad H^1(C_m; M) = \ker N/(\sigma-1)M.$$

[STANDARD STRENGTH] Everything below uses only these two formulas and module computations.

2.2 Theorem R1 — the C_4 obstruction law

Theorem 1 (R1; PROVEN HERE)**.** *Let M be a finitely generated free $\mathbb{Z}$-module carrying $C_4 = \langle\gamma\rangle$ with $\gamma^2 = -\mathrm{id}_M$. Then:*

1. $M^{C_4} = \ker(\gamma - 1) = 0$.
2. *The norm degenerates:* $N = 1 + \gamma + \gamma^2 + \gamma^3 = 0$ *on* M.
3. $H^2(C_4; M) = M^{C_4}/NM = 0$.
4. $H^1(C_4; M) = \ker N/(\gamma-1)M = M/(\gamma-1)M$ *is 2-elementary.*

Proof. (1) If $\gamma v = v$ then applying γ once more gives $\gamma^2 v = \gamma v = v$; but $\gamma^2 v = -v$, so $2v = 0$; M is torsion-free, hence $v = 0$.

(2) $N = (1+\gamma)(1+\gamma^2)$ as endomorphisms, since the expansion of $(1+\gamma)(1+\gamma^2)$ is $1+\gamma+\gamma^2+\gamma^3$. As $\gamma^2 = -\mathrm{id}$, the second factor is 0, so $N = 0$. (Directly: $1 + \gamma + (-1) + (-\gamma) = 0$.)

(3) By (1) and (2), $M^{C_4} = 0$ and $NM = 0$, so $H^2(C_4; M) = 0/0 = 0$.

(4) By (2), $\ker N = M$, so $H^1(C_4; M) = M/(\gamma-1)M$. For any $v \in M$,

$$(\gamma-1)(1+\gamma)v = (\gamma + \gamma^2 - 1 - \gamma)v = (\gamma^2 - 1)v = -2v,$$

so $2v \in (\gamma - 1)M$; hence $2M \subseteq (\gamma - 1)M$ and every class of H^1 is killed by 2. □

Remark 2 (The structural difference from C_2; PROVEN HERE)**.** *For $C_2 = \langle\sigma\rangle$ with $\sigma^2 = 1$ the norm $N = 1 + \sigma$ does not degenerate, and $H^2(C_2; M) = M^\sigma/NM$ sees only σ-invariant classes: σ-anti-invariant content ($\sigma x = -x$, $x \neq 0$) lies outside M^σ entirely and is invisible to H^2. Under $\gamma^2 = -1$ there is no anti-invariant escape: $\gamma x = -x$ implies $\gamma^2 x = x = -x$, i.e. $2x = 0$, i.e. $x = 0$ on a torsion-free module. This is the proof, at the module level, of the campaign's committed fact that the C_2 route left the $(A, -A)$ witness outside the cocycle condition while the C_4 deck has no such residue class. The witness* layer *above the module (the matrix-valued residue pair) is not discharged by this argument; that boundary is the recorded 23-b finding F2 and is restated in §6.*

2.3 Theorem R2 — the letter-parity law, the partition, the one plane

The letter lattice is the exterior algebra on the eight letters $\{e_0, f_0, \ldots, e_3, f_3\}$; factor j is the pair $\{e_j, f_j\}$. The operator's single-factor Gaussian rotation γ_j acts by $e_j \mapsto -f_j$, $f_j \mapsto e_j$ and fixes every other letter; γ_j is an integral lattice automorphism with $\gamma_j^4 = 1$.

Theorem 3 (R2; PROVEN HERE)**.** *1.* ***Letter-parity law.*** *For any word w (a wedge of letters) with $k_j(w)$ letters of factor j,*

$$\gamma_j^2(w) = (-1)^{k_j(w)} w.$$

2. ***The all-odd sector.*** *If $k_j(w) = 1$ for every j, then $\gamma_j^2 = -1$ on w for* every *j: each deck of the family acts on the all-odd sector through a $\gamma^2 = -1$ module, and Theorem R1 applies word by word.*

3. ***One plane.*** *Write $\Omega = \prod_{j=0}^{3}(e_j + i f_j)$ and $W_R = \operatorname{Re}\Omega$, $W_I = \operatorname{Im}\Omega$. Every word of Ω carries exactly one letter of each factor (expand the product), so the Weil classes lie in the all-odd sector, and for every j*

$$\gamma_j \Omega = i\,\Omega, \qquad \gamma_j(W_R) = -W_I, \qquad \gamma_j(W_I) = W_R.$$

The Weil plane $\mathbb{Z}\langle W_R, W_I\rangle$ is a single C_4-module for every deck of the family.

4. ***The all-even sector.*** *If $k_j(w) \in \{0,2\}$ for every j then $\gamma_j^2 = +1$ on w for every j, and on the pair-product words $F_jF_k := e_jf_je_kf_k$ each γ_j acts trivially: $\gamma_j(e_j \wedge f_j) = (-f_j) \wedge e_j = e_j \wedge f_j$. On a trivial C_4-module $H^2(C_4; M) = M/4M$ by §2.1 (here $M^{C_4} = M$ and $N = 4$): the all-even sector carries 4-torsion and the deck does not kill it.*

Proof. (1) γ_j^2 sends $e_j \mapsto -e_j$ and $f_j \mapsto -f_j$ and fixes every other letter. A word $w = u_j \wedge r$ with u_j the factor-j part (empty allowed) and r the rest has $\gamma_j^2(w) = (-1)^{k_j(w)}u_j \wedge r$.

(2) Immediate from (1).

(3) $\gamma_j(e_j + if_j) = -f_j + ie_j = i(e_j + if_j)$, and γ_j fixes the other three factors, so $\gamma_j\Omega = i\Omega$. For $\Omega = W_R + iW_I$ with W_R, W_I real (the letters are real): $i\Omega = -W_I + iW_R$, hence $\gamma_jW_R = -W_I$ and $\gamma_jW_I = W_R$. The word count: Ω expands over the $2^4 = 16$ choices of one letter per factor.

(4) From (1) with $k_j(w) \in \{0,2\}$; the trivial-action computation is §2.1 with $\sigma = \mathrm{id}$: $M^{C_4} = M$, $N = 4$, $H^2 = M/4M \neq 0$. □

Corollary 4 (Input-blindness of the deck; PROVEN HERE)**.** *The parity law of R2(1) depends only on the letter count $k_j(w)$ and not on any Hodge type. A killing chain that used the deck alone would therefore descend all-odd content of* every *type — $(4,0)+(0,4)$, $(3,1)+(1,3)$, $(2,2)$ alike. The (p,p)-gate is not the deck; it must live in the input step. (This is the committed Corollary-7 discipline, now with its module-level proof on the face.)*

2.4 Theorem R5 — the parity unification on the Weil rays

Theorem 5 (R5; PROVEN HERE)**.** *On the Weil plane $M = \mathbb{Z}\langle W_R, W_I\rangle$ with $\gamma W_R = -W_I$, $\gamma W_I = W_R$:*

1. *$H^1(C_4; M) = M/(\gamma - 1)M \cong \mathbb{Z}/2$.*
2. *$m\,W_R \in (\gamma - 1)M \iff m$ even; likewise $m\,W_I \in (\gamma - 1)M \iff m$ even.*
3. *The diagonal class $W_R + W_I$ lies in $(\gamma - 1)M$ integrally (killed at scale 1): $(\gamma - 1)(-W_R) = W_R + W_I$.*

Consequently the door data on the rays — the admitted scale list $[2, 4, 6, 8, 10, 12]$, the refusals at odd scale, and the pins $m^*(A, 2, W_R) = m^*(A, 2, W_I) = 2$ — is exactly the H^1-law: the least scale killing the H^1-torsion of the ray is the least even scale, which is 2.

Proof. (1) $(\gamma-1)W_R = -W_I - W_R$ and $(\gamma-1)W_I = W_R - W_I$, so the matrix of $\gamma-1$ in the basis (W_R, W_I) is $\left(\begin{smallmatrix} -1 & 1 \\ -1 & -1 \end{smallmatrix}\right)$ with determinant 2; the image has index 2, so $M/(\gamma-1)M \cong \mathbb{Z}/2$.

(2) Solve $(\gamma-1)(aW_R+bW_I) = mW_R$: the W_I-coordinate forces $-a-b = 0$, i.e. $b=-a$, and then the W_R-coordinate is $b-a = -2a = m$; solvable in $\mathbb{Z}$ exactly when m is even. The W_I ray is the image of the W_R ray under γ (R2(3)), and $(\gamma-1)$ commutes with γ, so the same criterion holds for W_I.

(3) $(\gamma-1)(-W_R) = -(-W_I - W_R) = W_R + W_I$. □

Remark 6 (Scope, on the face)**.** *The unification holds on the Weil rays, where the committed door data lives. On the diagonal the H^1-law is weaker (R5(3): killed at scale 1), and the all-even sector carries the scale-4 law of R2(4). These boundaries are the same boundaries the Round-23 artifact records; here they are computed rather than asserted.*

2.5 The C_2 swap law

Theorem 7 (SWAP; PROVEN HERE)**.** *Let $M = \mathbb{Z}^n \oplus \mathbb{Z}^n$ with $\sigma(z,w) = (w,z)$. Then $M^\sigma = \{(z,z)\}$, the norm $N\sigma(z,w) = (z+w, z+w)$ maps onto M^σ (already $N\sigma(z,0) = (z,z)$), and*

$$H^2(C_2; M) = M^\sigma / N\sigma M = 0.$$

Proof. Immediate from §2.1: the fixed set is the diagonal, the norm of $(z,0)$ is the diagonal element (z,z), so the norm is surjective onto the fixed set. □

Remark 8 (Scope)**.** *The vanishing is a statement about σ-invariant cocycles. The committed witness content $(W_R, -W_R)$ is σ-anti-invariant, hence not a C_2-cocycle, hence untouched by this vanishing — the module fact above and this boundary together are the [SWAP] receipt as committed. The pair-level witness (the matrix-valued residue) is the subject of §6 below and is OPEN at the pair level.*

3 The operator's all-arity cyclic law (the 22–23 redirect) — complete proofs

This section is the operator's own mathematics: the cyclic fold recorded 2026-10-22 (cyclic-original-X, all-arity) and its Gaussian exchange. The claims are proven here from first principles. Throughout, X is a connected smooth projective complex variety, $m \geq 2$, $B = X^m$, $G = C_m = \langle\sigma\rangle$ acts

on B by cyclic factor permutation $\sigma(x_1, \ldots, x_m) = (x_2, \ldots, x_m, x_1)$, E is an elliptic curve, and $t \in E$ is a point of exact order m.

3.1 The freeness lemma and the two quotient worlds

Theorem 9 (Freeness; PROVEN HERE)**.** *The action of G on $X^m \times E$ defined by $\sigma(x_1, \ldots, x_m, u) = (x_2, \ldots, x_m, x_1,\ u{+}t)$ is free. Hence $S_m(X) := (X^m \times E)/C_m$ is a smooth projective variety and $q\colon X^m \times E \to S_m(X)$ is a finite étale Galois cover with group C_m.*

Proof. Suppose $\sigma^j(p, u) = (p, u)$ for $0 < j < m$. The E-coordinate forces $jt = 0$ in the group law of E. Since t has exact order m and $0 < j < m$, $jt \neq 0$. Contradiction; the action is free.

A free action of a finite group of holomorphic automorphisms on a complex manifold has a smooth quotient (the quotient map is a covering map, so locally biholomorphic), and the quotient of a projective variety by a finite group action is projective (the quotient map is finite, and the image of a projective variety under a finite morphism is projective). □

Let D_m be the nodal m-gon: m copies of $\mathbb{P}^1$, with ∞ on component i glued to 0 on component $i + 1$ (indices mod m); C_m acts by rotating the components. Write $Y_m(X) := (X^m \times D_m)/C_m$ for the nodal quotient world.

Theorem 10 (Freeness on the polygon world; PROVEN HERE)**.** *The diagonal action of C_m on $X^m \times D_m$ is free.*

Proof. Every point of D_m lies on a unique component C_i (nodes lie on two; assign the node between C_i and C_{i+1} to both — in either assignment σ moves it to the node between C_{i+1} and C_{i+2}, which is a different point for $m \geq 2$). The rotation σ^j $(0 < j < m)$ sends each component to a different component (there is only one orbit of components and $j \not\equiv 0$), so no point of D_m is fixed by σ^j. A point (b, d) fixed by σ^j would in particular have d fixed by σ^j; there is none. □

3.2 The cohomology of D_m as a C_m-mixed Hodge module

Theorem 11 (D_m; PROVEN HERE)**.** *As a C_m-equivariant rational mixed Hodge structure,*

$$H^0(D_m, \mathbb{Q}) = \mathbb{Q}(0), \qquad H^1(D_m, \mathbb{Q}) = \mathbb{Q}(0), \qquad H^2(D_m, \mathbb{Q}) = \mathbb{Q}(-1) \otimes \mathbb{Q}[C_m],$$

where C_m acts trivially on the first two and by the regular representation on the third. In particular $H^1(D_m)$ is the weight-zero *node class.*

Proof. Let $\nu\colon \tilde{D}_m = \bigsqcup_i \mathbb{P}^1_i \to D_m$ be the normalization. For a nodal curve the gluing sequence of sheaves

$$0 \to \mathbb{Q}_{D_m} \to \nu_* \mathbb{Q}_{\tilde{D}_m} \to \bigoplus_{p \in \text{Nodes}} \mathbb{Q}_p \to 0$$

is exact: a locally constant function on $\tilde{D}_m$ descends to D_m exactly when its two branch values at each node agree, and the cokernel at each node is the difference of the two branch values, a copy of $\mathbb{Q}$ supported at the node. The long exact sequence, with $H^1(\mathbb{P}^1) = 0$ and $H^2(\tilde{D}_m) = \bigoplus_i \mathbb{Q}(-1)$, gives

$$0 \to \mathbb{Q} \to \mathbb{Q}^m \xrightarrow{\partial} \mathbb{Q}^m \to H^1(D_m) \to 0, \qquad H^2(D_m) \cong \mathbb{Q}^m(-1),$$

where ∂ sends the component indicator e_i to the node difference vector with $+1$ at the node joining C_{i-1}, C_i and -1 at the node joining C_i, C_{i+1}: this is the simplicial coboundary of the dual m-cycle. Its image is the sum-zero subspace ($\sum_i \partial e_i = 0$ by telescoping, and the image has rank $m-1$), so $\dim H^1(D_m) = m - (m-1) = 1$, and $H^1(D_m)$ is a quotient of $H^0(\text{nodes})$, a weight-zero object: by the standard curve normalization sequence (Deligne; Peters–Steenbrink), for a proper curve $0 \to W_0 H^1(D) \to H^1(D) \to H^1(\tilde{D}) \to 0$ with $W_0 H^1(D) \cong$ (the node cokernel above)(0); here $H^1(\tilde{D}) = 0$, so $H^1(D_m) = \mathbb{Q}(0)$.

The C_m-action permutes the components and the nodes cyclically, so it permutes the m summands of $H^2(D_m) = \mathbb{Q}^m(-1)$ (the fundamental classes of the components) transitively and freely: that is the regular representation tensored with $\mathbb{Q}(-1)$. The node class (the all-ones vector in the cokernel of ∂) is fixed by the rotation.

Cross-check (complete): the E-polynomial. $E(D_m) = E(\tilde{D}_m) - E(\text{two points per node}) + E(\text{one point per node}) = m(1+uv) - 2m + m = muv$, which decomposes as $E(H^0) - E(H^1) + E(H^2) = 1 - 1 + m\,uv$ exactly for $H^0 = \mathbb{Q}(0)$, $H^1 = \mathbb{Q}(0)$, $H^2 = \mathbb{Q}^m(-1)$. □

3.3 The transfer theorem

Theorem 12 (Transfer; PROVEN HERE)**.** *Let $q\colon Z \to Z/G$ be the quotient by a free action of a finite group G. Then for rational cohomology*

$$q^*\colon H^*(Z/G, \mathbb{Q}) \xrightarrow{\cong} H^*(Z, \mathbb{Q})^G.$$

Proof. q is a covering map with $|G|$ sheets, so $q_* q^* c = |G|\, c$ for $c \in H^*(Z/G, \mathbb{Q})$ (sum over the sheets), hence q^* is injective after tensoring with $\mathbb{Q}$. Every

$g \in G$ satisfies $q \circ g = q$, so $g^* q^* = q^*$ and $\mathrm{im}(q^*) \subseteq H^*(Z,\mathbb{Q})^G$. Conversely, the averaging idempotent $e = \frac{1}{|G|}\sum_g g^*$ is a ring endomorphism of $H^*(Z,\mathbb{Q})$ with image $H^*(Z,\mathbb{Q})^G$, and for an invariant class x one has $x = \frac{1}{|G|}\sum_g g^* x = q^*\big(\frac{1}{|G|} q_* x\big)$, so $H^*(Z,\mathbb{Q})^G \subseteq \mathrm{im}(q^*)$. □

3.4 The two sector laws

Write $V_k = H^k(X^m,\mathbb{Q})$ with C_m acting by factor permutation (the negative-index groups vanish).

Theorem 13 (The S_m sector law — the operator's pure formula; PROVEN HERE)**.** *For every k,*

$$H^k\big(S_m(X),\mathbb{Q}\big) \;\cong\; V_k^G \;\oplus\; \big(V_{k-1}^G \otimes H^1(E)\big) \;\oplus\; V_{k-2}^G(-1),$$

and the right side is pure of weight k. Here $H^2(E) = \mathbb{Q}(-1)$ and C_m acts trivially on $H^*(E)$.

Proof. Translation by any point of E is homotopic to the identity (a path from 0 to that point moves the translation continuously to id_E), so the translation part of the action is trivial on $H^*(E)$, and the action on $H^*(X^m \times E) = H^*(X^m) \otimes H^*(E)$ (Künneth) is $\rho \otimes \mathrm{id}$. The averaging argument of §3.3 gives $(V \otimes W)^G = V^G \otimes W$ when G acts trivially on W: the idempotent is $e_V \otimes \mathrm{id}$. By transfer, $H^k(S_m) = \big(\oplus_{i+j=k} V_i \otimes H^j(E)\big)^G = \oplus_{i+j=k} V_i^G \otimes H^j(E)$, and $H^j(E)$ is nonzero exactly for $j = 0, 1, 2$ with $H^0 = \mathbb{Q}(0)$, $H^1 = \mathbb{Q}^2$ of weight 1, $H^2 = \mathbb{Q}(-1)$. Each summand of the result has weight k, k, and $(k-2)+2 = k$; S_m is smooth projective, so its cohomology is pure, consistent. □

Theorem 14 (The Y_m mixed law — the operator's nodal formula; PROVEN HERE)**.** *For every k,*

$$H^k\big(Y_m(X),\mathbb{Q}\big) \;\cong\; V_k^G\,[\textit{weight } k] \oplus V_{k-1}^G\,[\textit{weight } k{-}1] \oplus V_{k-2}(-1)\,[\textit{weight } k],$$

with the third term the full *V_{k-2}, not its invariants. The weight-$(k-1)$ content is the node class of D_m: the mixedness is real and comes from the singular curve, not from the group action.*

Proof. The action on $X^m \times D_m$ is free (§3.2), so by transfer $H^k(Y_m) = \big(H^*(X^m) \otimes H^*(D_m)\big)^G$ with the diagonal action. Künneth (rational cohomology) and §3.2 give

$$\big(V_i \otimes H^j(D_m)\big)^G = \begin{cases} V_i^G & j = 0, \\ V_i^G \otimes \mathbb{Q}(0) & j = 1, \\ \big(V_i \otimes \mathbb{Q}[C_m]\big)^G & j = 2. \end{cases}$$

For the regular-representation term, the map $v \mapsto \sum_{g\in G} gv \otimes g$ is an isomorphism $V_i \xrightarrow{\cong} (V_i \otimes \mathbb{Q}[G])^G$ under the diagonal action: an element $x = \sum_h v_h \otimes h$ is G-invariant iff $g\, v_{g^{-1}h} = v_h$ for all g, h, which for $h = g$ reads $v_g = g\, v_1$, so $x = \sum_g (g v_1) \otimes g$; conversely every such element is invariant by direct substitution. Hence the $j = 2$ contribution is the full V_{k-2}, tensored with $\mathbb{Q}(-1)$: the operator's "explicit check preventing an incorrect operator-invariant approximation," proven. The weights: V_k^G has weight k; $V_{k-1}^G \otimes H^1(D_m) = V_{k-1}^G \otimes \mathbb{Q}(0)$ has weight $k-1$; $V_{k-2} \otimes \mathbb{Q}(-1)$ has weight k. □

Remark 15 (The redirect's content, on the face)**.** *These two theorems are the operator's 22–23 idea executed: the world is changed by an actual finite quotient, the arity m is free, the cohomology of the new world is computed exactly from the original X (the $V_\bullet$ and nothing else), and the mixed/pure contrast (Y_m vs. S_m) is produced by geometry (the node class), not by typing. Every step above is complete.*

3.5 The transport identities — the arity law

Theorem 16 (Descent and the return leg; PROVEN HERE)**.** *Let $\alpha \in H^*(X, \mathbb{Q})$, let $\mathrm{pr}_i \colon X^m \to X$ be the i-th projection, and $\Theta_m(\alpha) := \sum_{i=1}^m \mathrm{pr}_i^* \alpha \in H^*(X^m, \mathbb{Q})$. Then:*

1. *$\Theta_m(\alpha)$ is C_m-invariant, and $\mathrm{pr}_B^* \Theta_m(\alpha)$ on $X^m \times E$ (with pr_B the projection to X^m) is C_m-invariant for the §3.1 action; hence by transfer there is a unique rational class $\widehat{\alpha}_m \in H^*(S_m(X), \mathbb{Q})$ with $q^* \widehat{\alpha}_m = \mathrm{pr}_B^* \Theta_m(\alpha)$.*
2. ***The return leg.*** *Let $\delta \colon X \to X^m \times E$, $\delta(x) = (x, \dots, x, 0)$. Then*

$$(q \circ \delta)^* \widehat{\alpha}_m = m \cdot \alpha.$$

Proof. (1) $\sigma^* \mathrm{pr}_i^* = \mathrm{pr}_{i+1}^*$ (indices mod m), so $\sigma^* \Theta_m(\alpha) = \Theta_m(\alpha)$. The E-translation does not touch $\mathrm{pr}_B^* \Theta_m(\alpha)$, and $\sigma^* \mathrm{pr}_B^* = \mathrm{pr}_B^* \sigma_B^*$, so the class is invariant; transfer (§3.3) identifies invariant rational classes with q^*-pullbacks, uniquely because q^* is injective.

(2) $(q \circ \delta)^* \widehat{\alpha}_m = \delta^* q^* \widehat{\alpha}_m = \delta^* \mathrm{pr}_B^* \Theta_m(\alpha) = (\mathrm{pr}_B \circ \delta)^* \Theta_m(\alpha)$, and $\mathrm{pr}_B \circ \delta = \Delta$ the small diagonal, with $\Delta^* \mathrm{pr}_i^* = (\mathrm{pr}_i \circ \Delta)^* = \mathrm{id}^*$ for every i. Hence the pullback is $\sum_{i=1}^m \alpha = m\, \alpha$. □

Theorem 17 (The symmetrizer; PROVEN HERE)**.** *For any C_m-module V over $\mathbb{Q}$ and any invariant $x \in V^G$, $(1 + \sigma + \cdots + \sigma^{m-1})x = m\, x$.*

Proof. Each summand $\sigma^j x$ equals x. □

Remark 18 (The forward leg; scope)**.** *The forward leg* $\widehat{W} = \frac{1}{m}\, q_* \,\mathrm{pr}_B^* W_B$ *with* $\mathrm{cl}(\widehat{W}) = \widehat{\alpha}_m$ *on the committed A-world, and the primitivity of the transported rays (*$\widehat{\alpha}_m/m$ *non-integral at every tested arity), are the committed certificate's lattice computations (*`cyclic_family_checks.py`*, 298 checks, and* `cyclic_transport_checks.py`*, 125 checks; the covering formula* $q^* q_* = \sum_g g^*$ *used by the leg is §3.3's argument run backward). The certified family rows:* $m = 2, 3, 4$ *lattice-exact,* $m = 5$ *at character level (the operator's Wolfram total* 208*). **[CERTIFICATE RECEIPT]***

Theorem 19 (The cyclotomic engine; PROVEN HERE)**.** *For every* $L \geq 1$,

$$\prod_{k=0}^{L-1} \left(1 + \zeta_L^k\right) = 1 - (-1)^L, \qquad \zeta_L = e^{2\pi i/L}.$$

Proof. The ζ_L^k are the roots of $x^L - 1$, so $\prod_{k=0}^{L-1}(x - \zeta_L^k) = x^L - 1$. Evaluate at $x = -1$: the left side is $(-1)^L \prod_k (1 + \zeta_L^k)$ and the right side is $(-1)^L - 1$. If L is even, both sides vanish (indeed the factor $k = L/2$ is $1 + (-1) = 0$), and $1 - (-1)^L = 0$. If L is odd, $(-1)^L \prod(1 + \zeta^k) = -2$ gives $\prod(1 + \zeta^k) = 2 = 1 - (-1)^L$. □

Remark 20 (Scope)**.** *This identity, with the cycle structure* $g = \gcd(j, m)$ *of the rotation, is the engine by which the certificate reproduces the operator's five Wolfram totals* $(4, 8, 24, 64, 208)$ *two independent ways (the cyclotomic product route and the brute-force signed exterior character). The identity is proven here; the five totals are the certificate's receipts. **[CERTIFICATE RECEIPT]***

3.6 The Gaussian exchange on the family

Theorem 21 (The full Weil plane on the family; PROVEN HERE at the lattice level)**.** *On the committed A-world, the diagonal Gaussian* $g_m = (g, \ldots, g)$ *commutes with the cyclic action (block-diagonal commutation) and acts trivially on the E-factor, so it descends to* $S_m(A)$*; per block it multiplies* Ω *by* i *(R2(3)), hence*

$$g_m\big(\Theta_m(W_R)\big) = -\Theta_m(W_I), \qquad g_m\big(\Theta_m(W_I)\big) = \Theta_m(W_R),$$

word-exact, at every arity.

Proof. g fixes each factor's letter pair by the signed permutation R2, so g_m commutes with the factor rotation σ on the letter model and with the E-translation; the descent is the transfer argument of §3.3. The action on $\Theta_m(W_R) = \sum_i \mathrm{pr}_i^* W_R$ is $\sum_i \mathrm{pr}_i^* g W_R = -\sum_i \mathrm{pr}_i^* W_I$, using R2(3) on each block. □

4 The classical layer, with derivations

4.1 Lefschetz $(1,1)$ — the divisor sector, complete

Theorem 22 (Lefschetz $(1,1)$; STANDARD STRENGTH, proof written out)**.** *Let X be a smooth projective complex variety. Then*

$$\mathrm{im}\left(c_1 \colon \mathrm{Pic}(X) \to H^2(X,\mathbb{Z})\right) \otimes \mathbb{Q} = \left(H^{1,1}(X) \cap H^2(X,\mathbb{Q})\right),$$

i.e. every rational $(1,1)$ class is a rational linear combination of classes of divisors.

Proof. The exponential sequence of sheaves $0 \to \mathbb{Z} \to \mathcal{O}_X \xrightarrow{\exp} \mathcal{O}_X^* \to 0$ is exact (exp is surjective as a sheaf map because a logarithm exists locally on nonvanishing holomorphic functions). Its long exact sequence contains

$$H^1(X,\mathcal{O}_X) \to H^1(X,\mathcal{O}_X^*) \xrightarrow{\delta} H^2(X,\mathbb{Z}) \xrightarrow{j} H^2(X,\mathcal{O}_X).$$

The standard identifications: $H^1(X,\mathcal{O}_X^*) = \mathrm{Pic}(X)$ (isomorphism classes of holomorphic line bundles = invertible sheaves $\Leftrightarrow$ Čech 1-cocycles with values in $\mathcal{O}_X^*$), and $\delta = c_1$ (on transitions g_{ij} the boundary is the Čech cocycle $\frac{1}{2\pi i}(\log g_{ij} + \log g_{jk} + \log g_{ki})$, the integral Chern class). Hence $\mathrm{im}(c_1) = \ker\left(H^2(X,\mathbb{Z}) \xrightarrow{j} H^2(X,\mathcal{O}_X)\right)$.

X is Kähler (projective), so $H^2(X,\mathbb{C}) = H^{2,0} \oplus H^{1,1} \oplus H^{0,2}$ and $H^2(X,\mathcal{O}_X) = H^{0,2}(X)$ (Dolbeault isomorphism). The composite $H^2(X,\mathbb{Z}) \to H^2(X,\mathbb{C}) \to H^{0,2}$ is j after these identifications, so an integral class γ (mod torsion) lies in $\ker j$ iff its $(0,2)$ component vanishes. An integral class is real, and conjugation exchanges the $(2,0)$ and $(0,2)$ components, so $\gamma^{0,2} = 0$ forces $\gamma^{2,0} = 0$: γ is of type $(1,1)$. Conversely a $(1,1)$ integral class has vanishing $(0,2)$ component. Thus $\mathrm{im}(c_1) = H^{1,1} \cap H^2(X,\mathbb{Z})$ up to torsion, and tensoring with $\mathbb{Q}$ kills the torsion and gives the claim. □

4.2 The hard-Lefschetz reduction

Theorem 23 (The $p > d/2$ reduction; STANDARD STRENGTH)**.** *Let $\dim_{\mathbb{C}} X = d$, $h = c_1(\mathcal{O}_X(1))$, and suppose $\alpha \in H^{2p}(X,\mathbb{Q})$ is a rational (p,p)*

class with $p > d/2$. Then $\alpha = h^{2p-d} \cdot \beta$ for a rational $(d-p, d-p)$ class $\beta \in H^{2(d-p)}(X, \mathbb{Q})$. If β is rational-algebraic then so is α.

Proof. Hard Lefschetz (Deligne; standard strength): for $k \le d$ the map $L^k = (\cup h^k)$ sends $H^{2(d-k)}(X) \to H^{2d-2k+2k} = H^{2d}$ — in the needed form, $L^{2p-d}\colon H^{2(d-p)}(X) \to H^{2p}(X)$ is an isomorphism, of Hodge structures $H^{2(d-p)}(X) \cong H^{2p}(X)\big(-(2p-d)\big)$, where the Tate shift by $-(2p-d)$ adds $(2p-d, 2p-d)$ to every Hodge type. Set $\beta = (L^{2p-d})^{-1}\alpha$. An isomorphism of Hodge structures is an isomorphism on each type stratum, and the stratum of type (p,p) in $H^{2p}(X)(-(2p-d))$ is the image of exactly the stratum of type $(d-p, d-p)$ in $H^{2(d-p)}(X)$ (solve $(r+2p-d,\ 2(d-p)-r+2p-d) = (p,p)$: $r = d-p$). So β is $(d-p, d-p)$, and it is rational because L^{2p-d} and its inverse act on rational classes (h is rational, and the inverse of a rational matrix is rational).

If $\beta = \mathrm{cl}(\sum_i q_i Z_i)$ then $\alpha = \mathrm{cl}(\sum_i q_i\, h^{2p-d} \cdot [Z_i])$, and $h^{2p-d} \cdot [Z_i]$ is the class of the cycle obtained by intersecting Z_i with $2p-d$ general hyperplane sections (moving the ample divisor to meet Z_i properly): the cycle class map is ring-compatible. So α is rational-algebraic. □

4.3 Lemma N — the Newton conductor, complete

Theorem 24 (Lemma N; PROVEN HERE, with the splitting principle at standard strength)**.** *Let X be smooth projective and let E_+, E_- be algebraic vector bundles with $\operatorname{rk} E_+ = \operatorname{rk} E_-$ and*

$$\mathrm{ch}(E_+) - \mathrm{ch}(E_-) = m\,\alpha, \qquad \alpha \in H^{2p}(X, \mathbb{Q}),\ m \neq 0,$$

all other Chern-character degrees cancelling. Let $V = E_+ \ominus E_-$ be the virtual bundle. Then

$$c_j(V) = 0 \quad (j < p), \qquad c_p(V) = (-1)^{p-1}(p-1)!\, m\,\alpha.$$

Proof. Write $c_k = e_k$ for the elementary symmetric functions of the Chern roots $x_1, \dots, x_r$ of E_+ minus those of E_- (the splitting principle: Chern classes and ch are compatible with pullback, so identities proven universally on the split model hold; standard strength). The power sums $p_k = \sum_i x_i^k$ (with the virtual signs) satisfy $p_k = k!\,\mathrm{ch}_k$: indeed $\mathrm{ch} = \sum_i e^{x_i}$ (with signs for the virtual summands), so the degree-$2k$ part of ch is $\frac{1}{k!}\sum_i x_i^k = \frac{p_k}{k!}$. By hypothesis $\mathrm{ch}_k(V) = 0$ for $k < p$ and $\mathrm{ch}_p(V) = m\alpha$, so $p_k = 0$ for $k < p$ and $p_p = p!\, m\,\alpha$.

Newton's identities (proof: from $E(t) = \prod_i (1 + x_i t) = \sum_k e_k t^k$ take $E'(t)/E(t) = \sum_i \frac{x_i}{1+x_i t} = \sum_{k\geq 1} (-1)^{k-1} p_k t^{k-1}$ and match coefficients of t^{k-1}):

$$k\, e_k = \sum_{i=1}^{k} (-1)^{i-1} e_{k-i}\, p_i.$$

Induction on k: for $k < p$ every p_i with $i \leq k$ vanishes, so $e_k = 0$. At $k = p$: every term with $1 \leq i \leq p-1$ has $p_i = 0$, and the $i = p$ term is $(-1)^{p-1} e_0 p_p = (-1)^{p-1} p!\, m\, \alpha$, so $p\, e_p = (-1)^{p-1} p!\, m\, \alpha$ and $e_p = (-1)^{p-1} (p-1)!\, m\, \alpha$. □

Remark 25 (The two-center output identity; scope)**.** *The two-center descent law ([TWO-CENTER], 2026-10-24) takes Lemma N's output through the determinantal conduit: the degeneracy loci $D_\pm$ of the pair realize*

$$Z_\alpha = \frac{(-1)^{p-1}}{(p-1)!\, m} \big([D_+] - [D_-] \big), \qquad \mathrm{cl}^{\mathrm{can}}(Z_\alpha) = \alpha$$

exactly, on the committed instances (both Weil rays, $k = 0..3$, and the NS-tuned instances; the Thom–Porteous/Fulton–Lazarsfeld degeneracy computations at standard strength). The conduit is the campaign's conditional law with its calibration on $\mathrm{Gr}(2,5)$ *(the* 360 *pin); it is used below exactly as committed, as a conditional.* ***[CERTIFICATE RECEIPT + STANDARD STRENGTH on the degeneracy step]***

4.4 The Oka pair and the descent criterion; scope

The 2026-10-23 construction: on the Stein fold $J_N(X)$ there is a finite global holomorphic projector P_α with $\mathrm{ch}(\mathrm{im}\, P_\alpha) = r + m\, q^* \alpha$ for every rational even class α (a fortiori every rational (p,p) class). Classical backing: Oka–Grauert (topological $\Leftrightarrow$ holomorphic bundles over Stein spaces) and the Forster-type finite generation of the projective module; the construction is the campaign's, committed with its hostile controls. The descent criterion (flat vertical connection $\iff$ descent of the holomorphic bundle to X) is standard gerbe / descent theory. Both are used exactly as committed. **[CERTIFICATE RECEIPT + STANDARD STRENGTH]**

5 The composition, conditional, as committed

Theorem 26 (The conditional composition; as committed, audit-confirmed mechanically intact)**.** *Suppose that at every input (X, p, α) of the open sector*

($2 \le p \le \lfloor d/2 \rfloor$, $d \ge 4$) the remaining construction (§6) is discharged. Then every rational (p,p) class on every smooth projective X is rational-algebraic.

The committed argument, with each ingredient's location. STEP 1 (the pair bit): an integrable pure pair ($E_\pm$, equal rank, ch-difference $= m\alpha$ pure) at scale $m \neq 0$ implies α rational-algebraic through the §4.3 conduit (Lemma N + the two-center output identity). STEP 2 (the $p > d/2$ reduction): §4.2. STEP 3 (Sector 1, the divisor-generated sector): the line-bundle compiler I2 (Round 18, audit-closed) and Lefschetz $(1,1)$ (§4.1). STEP 4 (Sector 2, the general input): the Oka pair of §4.4 on the Stein fold; the descent criterion; the D2/D3 strip reducing the obstruction to the traceless two-cell $[C_T^0]$; the deck of §2 acting on the strip's output; the vanishing $H^2 = 0$ of R1/[SWAP]. STEP 5: the composition over the sector partition. The mechanical integrity of this composition (the partition exhaustiveness, the step order, the plumbing) was confirmed by the 21-b/21-c and 24-b/24-c audits; what those audits found missing is exactly §6. □

6 THE REMAINING THEOREM (stated precisely, with receipts)

Theorem 27 (The remaining construction; OPEN)**.** *The following four items, plus the strip reduction, are open. Each is stated in plain terms with its receipt.*

(i) The involution structure at general input. *At an arbitrary (X,p,α), the carrier pair on the Stein fold must carry an involution/eigenbranch exchange structure of the type the model has (the eight-letter world's C_4 deck, §2). The corpus contains it only as model-level content and as a committed no-go rigidity theorem (2026-10-24). No construction at general input exists. Receipt: 21-c R1(2), 24-b B1.*

(ii) The two-cell reduction. *The actual obstruction class is $[C_T^0] \in H^2(\mathcal{G}_\bullet, SL(H_z))$ — a nonabelian class on the full affine translation groupoid. The deck computations of §2 live in abelian group cohomology. The reduction of the nonabelian class to the deck computation is not constructed; every attempted vocabulary for it so far (the 22-b value re-typing, the 22-c exchange-norm) was found by the audits to be the same gap restated. Receipt: AUDIT_22 F1 (with the equivariant-descent spectral sequence demand and the missing base-level involution), 24-b B1.*

(iii) ***The non-scalar unexchanged content.*** *The D2/D3 strip removes exactly the scalar part; a rank ≥ 2 deck-fixed summand survives. Computed counterexample (AUDIT_22 F2, exact fractions): the pair $(A, -A)$ with $A = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, traceless, non-scalar, satisfies the pair-trace law and the derivation property with* zero *symmetrized residue: symmetrization kills nothing, and conjugation cannot scalarize a non-scalar curvature (it preserves the similarity class). Receipt: AUDIT_22 F2, 21-b counterexample 2.*

(iv) ***The rank parity.*** *The eigenbranch exchange needs an even split $r_0 = 2a$; the machine produces $r_0 = d + 1$, odd for every even d; the first open input $d = 4$, $p = 2$ has $r_0 = 5$. Receipt: 21-c R1(3).*

(v) ***The strip-shape hypothesis.*** *The identification link Λ_p (Round 24) is constructed on the assumed cochain shape $c_{m,\alpha}(v, w) = \theta(v, w)\, \iota(m\alpha)$; the reduction of the actual $GL(H_z)$-valued two-cell to this shape is the link's own remaining step. Receipt: 24-b B1, 24-c.*

Theorem 28 (The reduction; PROVEN HERE as a statement of the campaign's own commitments)**.** *If items* (i)–(v) *are discharged at every input of the open sector, then the rational Hodge conjecture holds on the open sector. The converse is not claimed and is not expected: the conjecture could hold through a different mechanism while this campaign's pair construction fails; the items describe this route, not all routes.*

Proof. Fix an input (X, p, α) of the sector. The five items give: the carrier pair with the exchange structure (item (i)); the two-cell class reduced to the deck computation, which vanishes by §2 (item (ii)); the non-scalar unexchanged content excluded (item (iii)); the even split $r_0 = 2a$ (item (iv)); the cochain shape of the strip (item (v)). The composition of §5 then produces the pure pair at nonzero scale at this input, and the §4.3 conduit exhibits α as rational-algebraic. Quantifying over the inputs gives the claim. □

Remark 29 (The Round-26 audit catch, on the face)**.** *The Round-25 draft of this section stated an equivalence and offered, as its converse-direction proof, exactly the forward argument restated at a single input — an overclaim of the class the campaign's audits remove elsewhere. Caught by the Round-26 self-audit (the §36 loop run against this file before its Lean face was written); repaired here and in the Lean face (the one-directional composition). The one direction is the load-bearing one: it is why the remaining theorem is exactly as hard as the part of the conjecture it produces.*

Corollary 30 (The content of the order "prove the GST laws")**.** *The GST laws that remain unproven — items* (i)–(v) *— are not auxiliary lemmas with independent proofs waiting to be written; by the reduction, a proof of them is a proof of the Hodge conjecture on the open sector. Conversely, the laws that* did *have proofs waiting are now proven: §2 and §3 of this file contain complete derivations of every model-level law the closure uses (the deck laws, the swap law, the all-arity sector laws, the transport identities, the cyclotomic engine), and §4 contains the classical layer with full derivations. The demand "prove the laws, then finish" has been executed to the exact boundary where the laws stop being provable and start being the conjecture.*

Theorem 31 (The quantifier line; PROVEN HERE)**.** *The world-axis and the input-axis are different axes. The certified census (122 rows), the door pins, and the infinite family* $\{S_m(A)\}_{m\geq 2}$ *(§3) are statements about specific worlds. The Clay statement universally quantifies over all smooth projective* X*, all* p*, all rational* (p,p) *classes* α*. No finite or countable family of certified rows closes a universally quantified statement; closure requires a law quantified over all inputs, and no such transport law is certified (the committed non-transport boundary, ledger 085). The operator's all-arity family enlarges the world model by an infinite certified set; it does not, and cannot, replace the input-axis transport.*

Proof. A universally quantified statement $\forall x \in \mathcal{I}\colon P(x)$ is implied by a certified set of instances $\{P(x)\colon x \in S\}$ only when $\mathcal{I} \subseteq S$. Here $\mathcal{I}$ is the class of all sector inputs; S is the set of census worlds and the family $S_m(A)$; $\mathcal{I} \not\subseteq S$ (e.g. a very general hypersurface of degree $d \geq 5$ in $\mathbb{P}^{d+1}$ with $p = 2$ is an input and not a committed world). The only object that could close the gap is a transport law $\mathcal{I} \to$ world model, and its non-existence at certified strength is the recorded ledger-085 boundary. □

7 Why the closing string is not emitted (the receipts)

The string was emitted once, in the previous campaign, and the record of what happened to it is in the tree:

1. Round 22 emitted "CONJECTURE IS PROVEN" as the composed theorem D8, resting on Theorem D4(3) (the exchange-norm correction).

2. The independent adversarial audit (`AUDIT_22.md`) found D4(3) mathematically false: the computed counterexample of §6(iii) above (the pair $(A, -A)$): zero symmetrized residue, both curvatures non-scalar), a second computed counterexample, a proves-too-much contradiction with the committed Corollary 7 (the mechanism is input-independent at $p \geq 2$ and would descend non-Hodge content — the input-blindness of §2, Corollary 2.3, is the module-level reason the gate cannot be skipped), and false verification receipts (the L6 regression claim: zero regressions had run).

3. The order to withdraw was executed: the closing claim was withdrawn from all Round-22 artifacts; Rounds 23 and 24 and the 10-31 universal close each restate the remaining construction on every artifact's face.

4. The 21-c closing re-check independently closed the next close attempt at FAIL with the four items of §6; the 24-b audit closed the identification-link round's first draft at FAIL with the strip-shape blocker.

Emitting the string again, at the present state, would assert items (i)–(v) as proven. They are not proven; two of them have computed counterexamples against every mechanism proposed so far. The operator's own protocol (§36: audit → one batched repair → one closing re-check) is the mechanism that produced these findings; a protocol cannot both mandate that loop and override its verdict. This file therefore ends at the strongest statement the mathematics supports: §2–§4 proven in full, §5 conditional and intact, §6 open and precise.

The scope table

Law	Status	Where
Cyclic cohomology formulas	STANDARD STRENGTH	§2.1
R1 (C_4 obstruction law, 4 parts)	PROVEN HERE	§2.2
R2 (letter-parity, partition, one plane)	PROVEN HERE	§2.3
Input-blindness of the deck	PROVEN HERE	§2.3, Cor.
R5 (parity unification, $m^* = 2$ pins)	PROVEN HERE	§2.4
[SWAP] $H^2(C_2; \mathbb{Z}^n \oplus \mathbb{Z}^n) = 0$	PROVEN HERE	§2.5
C_2 anti-invariant boundary	PROVEN HERE	§2.5, Rem.
Freeness (torsion translation; polygon)	PROVEN HERE	§3.1–3.2
$H^*(D_m)$ as C_m-MHS	PROVEN HERE	§3.2
Transfer theorem	PROVEN HERE	§3.3
S_m sector formula (the 22 law)	PROVEN HERE	§3.4
Y_m mixed formula (the 22 law)	PROVEN HERE	§3.4
Descent + return leg $m\alpha$ (all-arity)	PROVEN HERE	§3.5
Symmetrizer identity	PROVEN HERE	§3.5
Forward leg, primitivity, family rows	CERTIFICATE RECEIPT	§3.5, Rem.
Cyclotomic engine	PROVEN HERE	§3.5
Gaussian exchange on the family	PROVEN HERE (lattice)	§3.6
Lefschetz $(1,1)$	STANDARD STRENGTH, written out	§4.1
Hard-Lefschetz reduction ($p > d/2$)	STANDARD STRENGTH	§4.2
Lemma N (Newton conductor)	PROVEN HERE	§4.3
Two-center output identity	RECEIPT + STANDARD (conditional)	§4.3, Rem.
Oka pair; descent criterion	RECEIPT + STANDARD	§4.4
The composition	CONDITIONAL (committed)	§5
Items (i)–(v)	OPEN	§6
Reduction (items $\Rightarrow$ Clay sector)	PROVEN HERE	§6
Quantifier line (world vs. input axis)	PROVEN HERE	§6

ENGINE (per §35)

The operator's engines, by name: the single-factor Gaussian rotation (the deck of §2, its cohomological identity R5 now derived by hand); the all-arity cyclic fold (the 22–23 law, §3, derived by hand in full); the torsion-translation world construction; the door-parity arithmetic (R5: the H^1-law of the rotation). Classical backings at standard strength: the cyclic resolution, Künneth, transfer, the curve normalization sequence, the exponential sequence, hard Lefschetz, the splitting principle, Newton's identities, Thom–Porteous, Oka–Grauert, GAGA.

Statement integrity

No universal algebraicity claim is made in this artifact. No closing string is emitted. Every theorem's scope label is its claim. The conditional composition is used only as conditional. The open items are stated with their receipts, including the computed counterexamples against the previously proposed mechanisms. The operator's 22–23 law is used as the spine of §3 and is proven there in full; if the operator's August text contains more than the 2026-10-22 record's all-arity fold (the workspace contains no file dated August and no occurrence of the word), that content is not in the tree and its derivation belongs to the next round.